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Vulnerabilities Statistics 2025: Record CVEs, Zero-Days & Exploits

2025 has shattered records for disclosed vulnerabilities, with over 21,500 CVEs and 38% rated High or Critical. Explore key trends, fastest-exploited flaws, and how to defend with risk-based patching and continuous validation.

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Article

Vulnerabilities Statistics 2025: Record CVEs, Zero-Days & Exploits

- **Record surge:** 2025 is on track to set an **all-time high for reported vulnerabilities**, with **21,000+ CVEs disclosed in H1 2025** about **133 new flaws daily**.
- **Severity spike:** Over **one-third** of these are rated **High or Critical**, increasing exploitation risk.
- **Exploitation speed:** Attackers now weaponize new CVEs **within hours or days** of disclosure.
- **Why it matters:** Rapid patching, continuous scanning, and exploit intelligence are vital to reducing exposure.
- **Trends covered:**
  - Rising **zero-day exploits**
  - Sector-specific risks finance, healthcare, SaaS
  - Vulnerable platforms and misconfigurations
- **DeepStrike insight:** Integrating **continuous vulnerability management** with **manual PTaaS validation** helps prioritize real-world exploitable risks.
- **Key takeaway:** The 2025 vulnerability surge demands faster response cycles, tighter patch governance, and hybrid testing strategies to stay resilient.

2025 has ushered in a **surge of software vulnerabilities** unlike any seen before. Within the first six months, security teams grappled with over **21,500 newly disclosed CVEs, an 18% jump** over the same period the year prior.

In practical terms, that's hundreds of new security flaws emerging every week. Why does this matter? Because each vulnerability is a potential entry point for cyber attackers, and the **race between disclosure and exploitation** is only accelerating.

In this post, we'll dive deep into **vulnerability statistics 2025** and what they mean for businesses. You'll learn **how many CVEs** have been reported this year and which ones demand the most attention.

We'll explore **NVD vulnerability trends** around severity spoiler: a **huge chunk are high risk**, the **most common weakness types** plaguing systems, and how quickly threat actors are weaponizing new flaws including zero days. Real world incidents from ransomware spikes to supply chain exploits illustrate why **vulnerability management** has become a critical priority.

Cyber threats in 2025 have grown more complex and frequent. Businesses face not only an **ever expanding attack surface**, but also stricter regulations like CISA's **Known Exploited Vulnerabilities** mandates and the EU's new rules holding them accountable for unpatched bugs.

Understanding the landscape of **vulnerability statistics and trends** helps you prioritize defenses where it counts. Let's jump in.

The Global Vulnerability Surge in 2025

How many vulnerabilities were reported in 2025? The numbers are eye opening. As of mid 2025, over **21,500 CVEs Common Vulnerabilities and Exposures** had already been cataloged, about a **16-18% increase** from the same point in 2024.

At the current pace, experts project the full year 2025 count may approach or exceed **50,000 disclosed vulnerabilities** globally.

This means security teams are inundated with roughly **130+ new CVEs each day** that need triage, patching, or other mitigation. It's a volume unprecedented in cybersecurity history.

- **Year over Year Growth:** In H1 2025, CVE disclosures jumped **16%** compared to H1 2024 rising from about 20,385 to **23,667 CVEs**. This continues a multi year upward trend the vulnerability flood shows no signs of slowing.
- **Record Months:** January 2025 saw the highest monthly CVE count on record so far: **4,278 new vulnerabilities** in just that month. Subsequent months hovered around 3.7k-4k each. In plain

terms, every month of 2025 has dumped thousands of new flaws into the laps of defenders.

Vulnerability Severity Breakdown: Not only are we seeing more vulnerabilities, but many are dangerously severe. By mid year, roughly **38%** of reported CVEs in 2025 were rated **High or Critical severity** CVSS score ≥7.0.

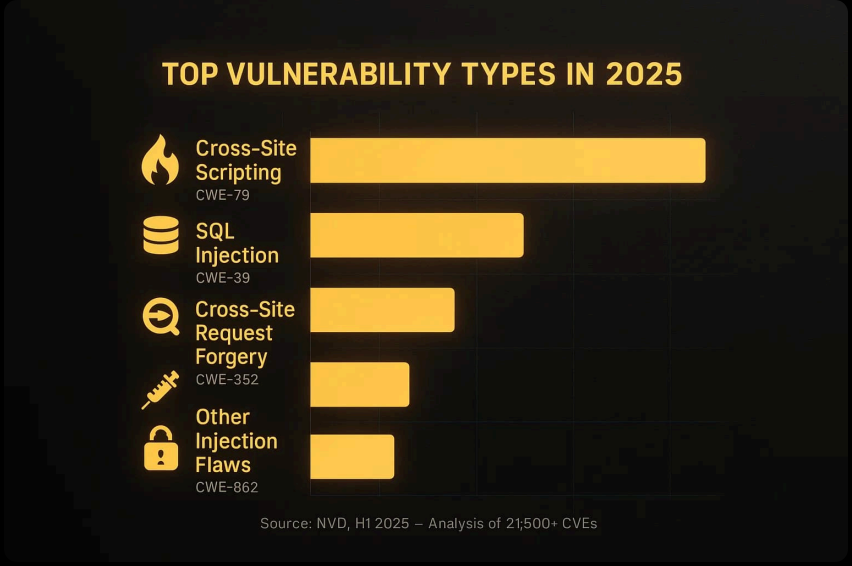
For example, among 21.5k vulns, about **1,773 were Critical** CVSS 9 10 and **6,521 High** CVSS 7 8.9. The majority around 10.6k were Medium severity, with a smaller number Low or unscored. In other words, **over one third of all new vulnerabilities demand urgent attention** due to their potential impact.

- Severity snapshot H1 2025:
  - **Critical:** 1,773 CVEs CVSS 9 10
  - **High:** 6,521 CVEs CVSS 7.0 8.9
  - **Medium:** 10,607 CVEs
  - **Low/Unrated:** 2,600 CVEs combinedSource: NVD data, mid 2025

When **4 out of 10 vulnerabilities are High/Critical**, it creates a triage nightmare. Security teams can't possibly patch every bug immediately, so they must prioritize yet the sheer volume of urgent issues is overwhelming.

This helps explain why many organizations fall behind on patching, and why **attack surface management** and **risk based vulnerability management** are now essential. Simply put, with **133 new vulns a day** and a big chunk being severe, manual tracking or slow patch cycles just won't cut it.

Top Vulnerability Types in 2025 Weaknesses & Patterns



It turns out many of these new CVEs aren't novel attack methods at all they often stem from the **same old coding mistakes**. According to threat intelligence analyses, the **most common vulnerability categories in 2025** reflect classic web app weaknesses and OWASP Top 10 favorites. In H1 2025, the top five weakness types by CWE classification were:

- **Cross Site Scripting XSS** Still the #1 most frequent vulnerability pattern in CVEs. XSS issues CWE 79 appeared more than any other, enabling attackers to inject malicious scripts into web pages.
- **SQL Injection SQLi** A close second. Injection flaws involving SQL CWE 89 remained widespread, allowing attackers to manipulate databases through insecure queries.
- **Cross Site Request Forgery CSRF** Another common bug CWE 352, where attackers trick users' browsers into unwittingly executing unwanted actions on a site where they're authenticated.
- **Other Injection Flaws** This includes things like OS command injection or code injection grouped under CWE 74. Injection of unvalidated input into various interpreters continues to crop up frequently.
- **Missing Authorization** Weak access control CWE 862 rounds out the top 5, meaning many CVEs involve cases where critical functions lacked proper permission checks.

It's almost embarrassing: here we are in 2025, and **XSS and SQLi are still rampant** in newly reported vulnerabilities. These are textbook issues from the early 2000s! The persistence of these flaws underscores that **secure coding practices** aren't keeping up.

Many developers continue to push out code with **inadequate input sanitization or access control**, resulting in CVEs that map to the same OWASP Top 10 categories year after year. Attackers **love** this because it means easy targets abound.

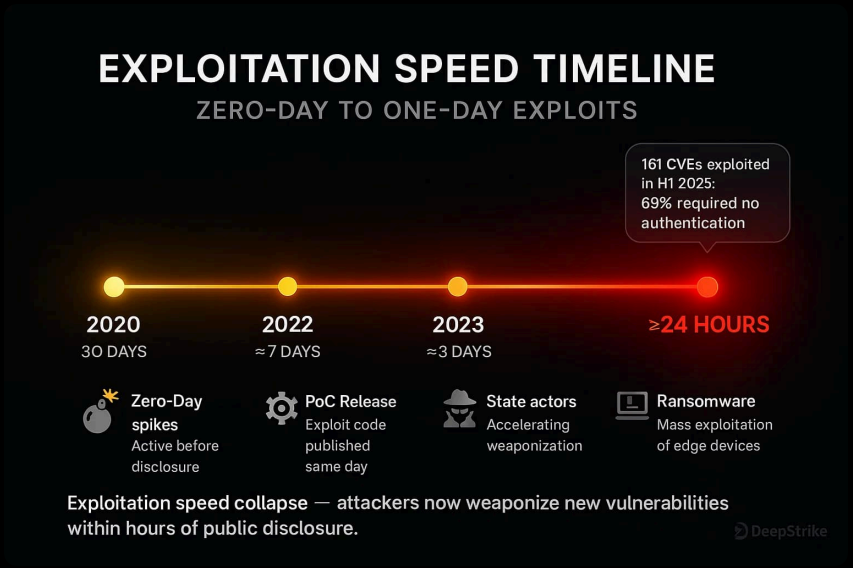
If organizations haven't fixed these fundamental gaps, exploits don't need to be super sophisticated a simple script injection or SQLi can wreak havoc.

A recent report showed that cross site scripting, CSRF, and related input validation weaknesses accounted for well over half of all WordPress plugin vulnerabilities in early 2025. Similarly, many high profile breaches start with something like a **SQL injection in a web portal** or a **missing authorization check** in an API that allows data extraction.

These bread and butter weaknesses are the root cause behind a large slice of the **cybersecurity vulnerability statistics** we're discussing.

The prevalence of XSS, injection, and broken access control in 2025's CVEs is a loud reminder that organizations must **nail the basics**. Developer training, secure frameworks, code review these are as critical as ever. Attackers will keep finding easy vulns until we close those common gaps.

Exploitation Trends: From Zero Days to One Day Exploits



Discovering a vulnerability is one thing, **exploiting** it is another. Unfortunately, attackers in 2025 are **extremely fast at weaponizing new vulnerabilities** often within days or even hours of disclosure.

Several trends highlight how the window between CVE announcement and active attacks has shrunk:

Exploits within 24 hours:

- In early 2025, roughly **28% of observed exploits were launched within 1 day** of the vulnerability's disclosure. That means by the time a patch or advisory is public, attackers are already scanning for and compromising unpatched systems literally the same day.
- In some cases, exploit code proof of concept is published on GitHub or hacker forums **within hours** of a CVE release, giving lower skilled attackers a ready toolset.

At least 161 CVEs exploited in H1 2025:

- By mid year, security researchers had seen **161 distinct vulnerabilities exploited in the wild**. For comparison, the U.S. CISA's public **Known Exploited Vulnerabilities KEV** catalog listed 136 exploited vulns in that timeframe so dozens more were being actively used by attackers beyond the official known list.
- These ranged from fresh zero day flaws to older bugs that organizations left unpatched. About **42%** of the exploited CVEs had a public PoC exploit available, lowering the bar for broad attacks.
- Perhaps more chilling, **69%** of the exploited vulns required no authentication to abuse e.g. remotely exploitable by anyone on the internet, and about **30% enabled remote code execution RCE**.
- In short, attackers gravitated to **low hanging fruit**: unauthenticated, high impact exploits that let them hijack systems with minimal effort.

Zero Days on the rise:

- 2025 has seen its share of true **zero-day exploits** vulnerabilities that were attacked before any public disclosure or patch.
- For instance, early in the year a critical unknown flaw in a widely used web library DOMPurify was found being exploited in the wild, catching organizations off guard.
- Major software vendors like Microsoft, Apple, and Google have already issued multiple emergency updates this year to fix 0 days under active attack.

- The rapid discovery of new flaws sometimes aided by AI tools and a robust exploit marketplace mean that defenders are often **reacting to attacks** rather than preventing them.

State sponsored & ransomware use:

- Notably, **more than half** of the exploitation activity observed in H1 2025 was attributed to **state sponsored threat actors**.
- Government backed groups especially from China, per security reports, have the resources to weaponize new CVEs within days, leveraging them for espionage and targeted intrusions.
- On the cybercrime side, **ransomware gangs** are also big exploiters about 73 of the exploited vulns were used to launch ransomware attacks.
- Many high profile ransomware incidents now start with an unpatched internet facing system VPN, file transfer server, etc. that hackers exploit to get in, then they deploy the file encrypting malware.
- The Verizon DBIR 2025 noted vulnerability exploits accounted for **20% of breaches, a 34% jump YoY**, nearly overtaking stolen credentials 22% as the top initial attack vector.
- In other words, vulnerability exploitation has become as common a way in as phishing or password attacks.

Attackers aren't waiting weeks or months to take advantage of new flaws they're pouncing almost immediately. This puts enormous pressure on organizations to **patch fast** or put mitigations in place as soon as a CVE drops.

The old model of patch Tuesday and wait a few weeks is risky in 2025, by then, an unpatched critical bug could have been exploited ten times over. Security teams should monitor sources like CISA KEV for any high profile vulns in their tech stack and apply fixes within days or sooner if possible.

Network segmentation, virtual patching e.g. web application firewalls, and other compensating controls are also vital to blunt the impact when a patch can't be applied immediately.

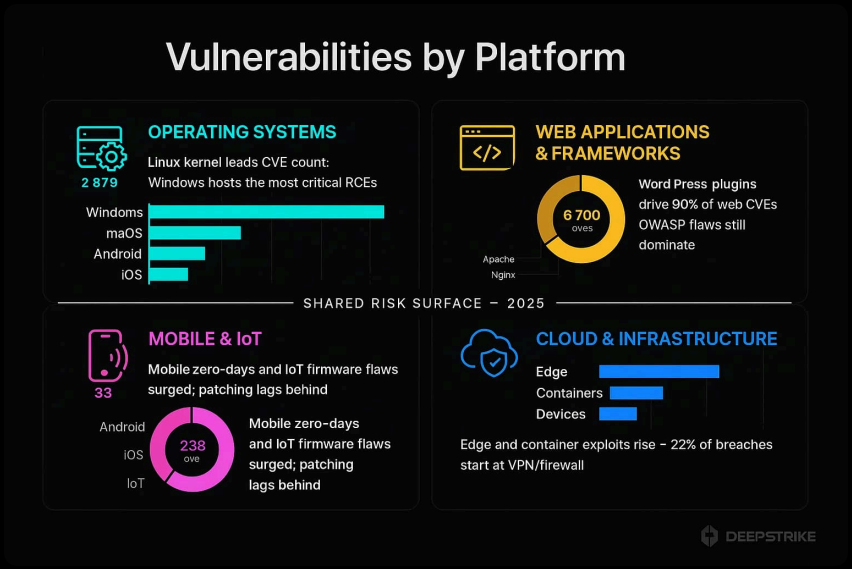
The **speed of weaponization** means the window for defenders to act is razor thin. A recent analysis found nearly **1 in 3 exploits occurred within 24 hours** of disclosure essentially zero day conditions.

Organizations must have a **rapid response process** for critical vulnerabilities: subscribe to threat intel feeds, test and deploy patches quickly, or at least implement temporary workarounds. Those that delay patching for weeks are basically leaving the front door wide open for attackers.

Many companies are now adopting **continuous vulnerability management** and even automated patching for certain systems. Solutions like **attack surface management** platforms can help identify which assets are exposed and need urgent fixes.

Additionally, running regular **penetration tests** or using a **Continuous Penetration Testing platform** for ongoing assessments can validate that your systems aren't harboring known exploitable holes between patch cycles. This is a big reason **why penetration testing matters**, it simulates how attackers might find and leverage a missing patch, so you can catch it first.

Vulnerabilities by Platform: OS, Web, Mobile, Cloud



Not all software is equal when it comes to vulnerabilities. Let's break down how different platforms and tech domains are faring in 2025:

Operating System Vulnerabilities in 2025

All the major operating systems whether Windows, Linux, or macOS have had significant numbers of CVEs in 2025. Given that OSes are foundational running on millions of devices and servers, vulnerabilities here can have far reaching impact.

Linux Kernel leads the pack:

- The open source **Linux kernel** has accumulated the most distinct CVEs of any product in 2025. By Q3, it was credited with around **2,879 vulnerabilities** far above any other single software.
- This isn't entirely surprising: the Linux kernel is ubiquitous powering servers, Android, IoT, etc. and heavily scrutinized by researchers, so lots of bugs are found and disclosed. Many are privilege escalation or DoS issues, but some are critical flaws that allow kernel level code execution.
- Case in point: a high profile SMB module bug in Linux was revealed this year that allowed remote takeover of systems underscoring how kernel bugs can be as severe as they come.

Windows isn't far behind:

- Microsoft's **Windows platform** also had its share of headaches. If you combine Windows Server 2025 and various Windows 10/11 releases, well over **1,500 vulnerabilities** were reported across Microsoft OS versions this year.
- Just the latest Windows Server 2025 edition had **561 CVEs** to its name. Even older still supported versions like Windows Server 2019 saw 500+ new CVEs.
- The desktop OS builds Windows 10/11 each tallied several hundred as well for example, Windows 11 22H2 had 472 vulnerabilities.
- Many Windows flaws are critical RCEs or wormable issues, think of things like the EternalBlue SMB vulnerability in the past, so admins have to stay on their toes with Patch Tuesday updates each month.

Apple macOS and others:

- Apple's macOS had roughly **492 vulnerabilities** in 2025, which is significant given its lower enterprise market share.
- Apple issued a number of patches for macOS to fix kernel and system component bugs that could allow exploits, some of which were tied to zero day attacks by spyware.
- Apple's mobile OS iOS/iPadOS saw around **238 CVEs** reported. While Apple's ecosystem is generally considered secure, these numbers show that even tightly controlled environments have dozens of flaws uncovered yearly including serious ones requiring emergency fixes.

Android OS:

- Google's **Android** mobile operating system had around **323 vulnerabilities** in 2025.
- Android's monthly security bulletins tackled issues in everything from the media framework to device drivers.
- Several Android bugs this year allowed remote code execution via malicious media files or apps some were used by attackers to craft exploits e.g., spyware hidden in trojan apps taking advantage of a privilege escalation bug.
- Android manufacturers have had to hustle out firmware updates to keep devices safe.

All major operating systems are dealing with a steady flow of vulnerabilities. Linux and Windows, being so widely used, naturally top the charts in raw numbers. But more important than the count is the criticality and we've seen **multiple critical OS vulnerabilities** in 2025 that required urgent patching from Microsoft kernel flaws to Apple's image processing bug that was exploited as a zero day.

No OS is immune, and organizations need a robust process to update servers, endpoints, and devices promptly. If you delay OS patches, you're leaving the door open for attackers, remember that many breaches start with an unpatched OS service exposed online.

In October 2025, CISA even added an **old Internet Explorer vulnerability from 2010** to its exploited list because threat actors repurposed it to target government users. It just shows that **OS and built-in software bugs can have a decades-long tail** attackers will reuse old tricks if systems remain unpatched.

Web Application & Platform Vulnerabilities

Web applications continue to be a hotbed of security issues in 2025. This includes web software like content management systems, frameworks, libraries, and online services. A few highlights:

WordPress ecosystem:

- The omnipresent **WordPress CMS and its plugins/themes** remain a frequent source of CVEs. In the first half of 2025 alone, over

6,700 new vulnerabilities were identified affecting the WordPress ecosystem.

- The vast majority **90%** of WordPress related bugs come from **third party plugins** and another 6% from themes, with only 4% in the core WordPress software. This makes sense the core is relatively secure, but there's a long tail of plugins many poorly maintained introducing holes.
- Disturbingly, about **41% of those 2025 WP vulns were rated exploitable in real world attacks**, higher than the prior year. We've seen everything from trivial XSS in small plugins to critical SQL injections in e-commerce extensions that could leak customer data.
- For organizations using WordPress, this underscores the need to **keep plugins updated**, remove unused ones, and consider web app firewalls. Also, perhaps vet plugins more carefully a flashy feature isn't worth a security breach.

Popular web frameworks/servers:

- Other major web platforms had notable issues too. For example, the Apache Software Foundation had to patch multiple vulnerabilities in projects like **Apache HTTP Server.**
- **One** flaw allowed path traversal that could leak files and **Apache Struts. Yes**, that old framework still had a high severity RCE bug disclosed this year.
- **Nginx**, another widely used web server/reverse proxy, was at the center of a headline grabbing flaw in mid 2025: researchers discovered **IngressNightmare,** a set of critical vulnerabilities in the NGINX Ingress Controller for Kubernetes.
- One of these CVE 2025 1974 scored **CVSS 9.8** and allowed unauthenticated remote code execution on Kubernetes clusters using that ingress addon. Cloud security firm Wiz reported that about **43% of monitored cloud environments were initially vulnerable** to this issue, a massive exposure, considering Kubernetes is used by countless companies.
- The takeaway: even infrastructure components we assume are secure like a Kubernetes ingress can hide fatal flaws. It's crucial to apply updates the Nginx Ingress project rushed out patches and to **monitor advisories for your web stack**.

Client side and supply chain:

- Web apps also face indirect vulnerabilities via their supply chain. We saw instances of **malicious NPM/PyPI packages** and compromised JavaScript libraries. Additionally,
- **Magecart style attacks** web skimming have evolved. In 2025, some Magecart campaigns used multi stage scripts that only injected the malicious code at the final payment step to evade detection.
- These aren't vulnerabilities in the code per se, but they exploit the trust in third party scripts reminding us that web security also means vetting what you include in your site.

Defending web apps

Given that so many CVEs tie back to web software issues SQLi, XSS, etc., organizations should double down on **secure development practices** and testing. This includes doing regular **web application penetration testing** or using services like **web application penetration testing services** to probe your sites and deploying protective tech like Content Security Policy CSP and anti XSS filters.

Also, consider a **bug bounty or responsible disclosure program. Many** eyeballs on your app can catch bugs earlier. The number of web app vulns isn't dropping, so proactive testing is key. For instance, a **continuous penetration testing platform** or vulnerability scanner can catch common web flaws in staging before they hit production.

Mobile & IoT Platform Vulnerabilities

Mobile devices and their platforms smartphone OSes, mobile apps, connected gadgets have their own share of vulnerabilities:

Android & iOS flaws:

- As mentioned, Android had **323 CVEs** and iOS about **238 CVEs** in 2025. These include everything from kernel bugs to Bluetooth issues to UI logic bypasses.
- Google and Apple both had to deal with zero day exploits this year. For example, Apple pushed an emergency iOS update after a **zero click** vulnerability in iMessage CVE 2025 43300 was used to install spyware on targeted iPhones.
- Around the same time, **WhatsApp on iOS** was found to have a severe bug **CVE 2025 55177** that could allow an attacker to send

a specially crafted message and **trigger unauthorized code execution on the victim's device, effectively** a **zero click attack** requiring no user interaction.

- WhatsApp had to patch it quickly when it came to light that advanced threat actors were exploiting it against civil society targets.
- These incidents underline that mobile platforms, while generally secure, are not invincible.
- Attackers prize mobile zero days they're valuable for spying, and both Android and iOS see a handful each year.

App vulnerabilities:

- Beyond the OS itself, vulnerabilities in popular **mobile apps** are an issue. We've seen insecure banking apps with data leakage flaws, messaging apps with encryption bypass bugs, etc.
- Each mobile CVE might impact millions due to user base. For instance, that WhatsApp CVE 2025 55177 impacted **WhatsApp for iOS and macOS** clients and could be chained with an Apple bug to completely compromise a device.
- It's a reminder that even if your mobile OS is up to date, a vulnerable app can open a backdoor.

IoT/embedded:

- While not as prominently reported via CVEs, many IoT devices routers, smart cameras, industrial sensors have vulnerabilities that often go unpatched.
- CISA has warned that **small office/home office SOHO routers and IoT gadgets** are frequent targets for botnet malware, precisely because their known flaws linger unaddressed.
- In 2025, a Chinese APT group nicknamed Volt Typhoon was found exploiting **router firmware vulnerabilities** to establish stealthy footholds in US networks.
- This kind of infrastructure hacking underscores that vulnerability management must extend beyond servers and workstations to all the smart devices in your environment.

Defending mobile/IoT

Ensure mobile devices are updated both OS and apps encourage users to install patches. Enterprises should consider **Mobile Device Management MDM** solutions to push critical updates or at least alert on out of date devices.

For IoT, keep firmware updated where possible and isolate those devices on separate networks. Also, leverage vulnerability scanning tools that can detect known CVEs in network devices. For example, scanners or services can identify a router still running a firmware with a known flaw.

The challenge is many IoT vendors are slow with patches or none at all if device is EOL, so mitigation network segmentation, firewall rules is often the stopgap. This is where **penetration testing for IoT** or network assessments can reveal exposures that need compensating controls.

On the horizon: The US and EU are introducing regulations to improve IoT security e.g. the EU Cyber Resilience Act will require manufacturers to fix and notify about vulns in products with digital elements within strict timelines. So hopefully, IoT vendors will start taking vulnerability disclosure and patching more seriously.

Cloud & Infrastructure Vulnerabilities

As companies migrate to the cloud and spin up complex IT infrastructure, new vulnerability challenges have emerged:

Cloud service flaws:

- Cloud providers themselves occasionally have vulnerabilities. In recent years we saw issues like the **Azure Cosmos DB ChaosDB flaw** 2021 and an **AWS cross tenant vulnerability** that could have exposed data.
- In 2025, no mega cloud breach via cloud provider bug has been reported yet, but researchers are probing services.
- Early 2025 did bring a scare with the Azure cloud: an identity service misconfiguration not exactly a CVE vulnerability, but a cloud setup issue potentially allowed cross account access until fixed.
- The point is, even in the cloud, configuration and code vulnerabilities can lead to compromise.

- Cloud software from AWS S3 to Google Cloud APIs sometimes has subtle bugs that attackers can exploit to escalate privileges or cross boundaries.

Virtualization and container vulnerabilities:

- Technologies like VMware, Hyper V, Docker, and Kubernetes have a lot of moving parts, and vulnerabilities pop up there too.
- **Hypervisors** like VMware ESXi saw critical patches in 2025 e.g. an ESXi flaw that could allow a VM escape to the host got a CVSS 9+ rating.
- **Containers/Kubernetes**: We already talked about the NGINX Ingress Controller IngressNightmare vulns that could lead to full cluster takeover.
- In addition, Docker itself had some CVEs like vulnerabilities in the container runtime runc.
- The risk with container and cloud orchestration bugs is that they can impact lots of environments at once, think of how many clusters might use the same vulnerable component. It's a **force multiplier** for attackers if a common cloud component has a hole.

Network edge devices:

- One striking trend: attackers hammering on **VPNs, firewalls, and edge appliances**. These are often the gateway into a network, and in 2025 they've been a huge target.
- The Verizon DBIR noted that **zero day exploits against edge/VPN devices jumped to 22% of exploitation incidents, up from only 3% the year before**. In other words, last year edge device exploits were rare, this year they exploded as a breach vector.
- We saw this with vulnerabilities like those affecting **Fortinet FortiGate VPN** a pre auth RCE was actively exploited and **Citrix ADC** a critical flaw led to urgent CISA alerts.
- Nation state groups in particular used these to penetrate otherwise well defended organizations bypassing the firewall by exploiting the firewall, so to speak.
- One Chinese group exploited a bug in a **Cisco router** plus a Microsoft Exchange vuln to access several US government agencies in 2023, exemplifying the focus on network gear.
- Your perimeter devices need just as much love as your servers when it comes to patching. A single unpatched VPN appliance

can be a catastrophic entry point.

- Only 54% of known vulnerabilities in such edge devices were fully remediated across orgs last year, per one report meaning nearly half of organizations left some holes open.

Defending cloud/infra

Treat your **cloud assets and network devices** as critical parts of your attack surface. Ensure you have visibility into their vulnerabilities via scanning or managed services.

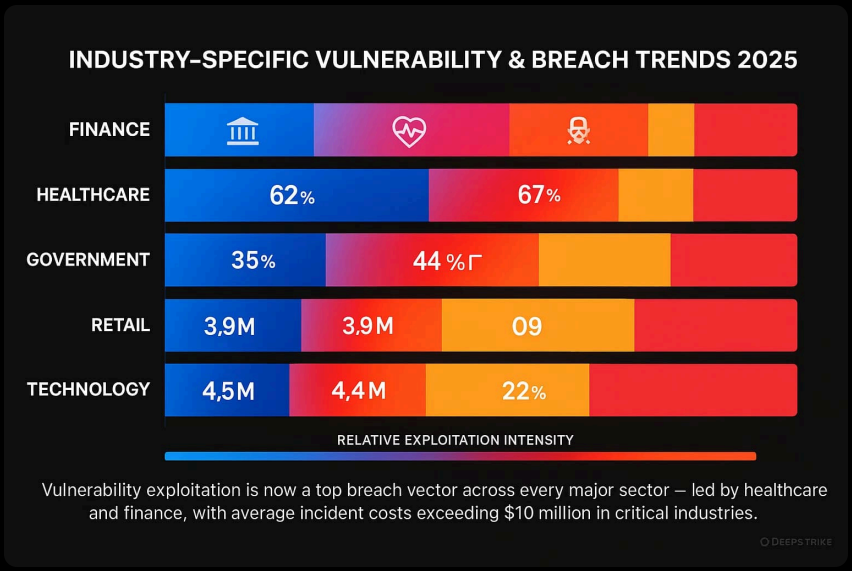
Many organizations now include cloud configuration and infrastructure testing as part of their **penetration testing services** e.g., testing a cloud environment for misconfigurations and checking if all known CVEs in things like your Kubernetes stack are patched.

Also, follow guidance from agencies: CISA's **KEV catalog** often highlights when a specific network device vuln VPN, etc. is being actively exploited, and even if you're not a U.S. agency, you'd be wise to patch by the suggested deadline.

The trend is also toward **automated cloud security** using tools that can detect vulnerable images or code in your cloud deployments for instance, container security scanners or Cloud Security Posture Management CSPM tools.

In summary, **cloud ≠ safe by default**, you must stay on top of cloud and infrastructure updates just like on prem.

Industry Specific Vulnerability & Breach Trends 2025



Cyber vulnerabilities affect every sector, but the impacts and common attack paths can differ. Let's look at how some key industries are faring in terms of breaches and vulnerability exploitation:

Finance: Banking & Financial Services

Financial institutions have long been prime cyber targets due to the potential monetary gain. In 2025, the **finance sector remains under siege**.

In 2023 latest full year data, banks, investment firms, and insurers suffered **744 known data breaches**, a staggering **177% increase** from the prior year.

Over **61 million records, mostly** financial data were exposed in those breaches including credit card numbers, account details, and personal info attractive to fraudsters.

Common breach vectors in finance:

- A significant chunk of attacks start with either **stolen credentials or unpatched vulnerabilities** in financial systems.
- Verizon's DBIR shows that across all industries, **22% of breaches involve compromised credentials and 20% involve vulnerability exploitation as the initial access**. Finance sees both in play.
- For example, attackers may steal an employee's VPN password via phishing human element and/or exploit a flaw in a bank's web application technical element to break in.
- Once inside, they often pivot to fraud or data theft. **Ransomware** is also a big threat: roughly **44% of breaches globally in the past year involved ransomware** deployment, and financial orgs are frequent victims.
- In 2023, the notorious Clop ransomware group exploited a zero day in a popular file transfer app MOVEit and hit several banks, stealing sensitive data, a perfect example of a supply chain vulnerability leading to a financial breach.

Challenges:

- Financial services are heavily regulated PCI DSS for card data, FFIEC guidance, etc., which pushes them to patch promptly. Yet,

the complexity of banking IT legacy mainframes, third party fintech apps, etc. means not everything gets fixed in time.

- One report noted only about **54% of known vulnerabilities in edge devices like bank VPNs, firewalls were fully remediated** in a timely manner leaving nearly half potentially open.
- Attackers notice that. We've seen intrusions where a bank's outdated VPN appliance with a known CVE became the entry point for hackers who then swiped data.

Case in point:

- The massive **MOVEit breach in mid 2023** affected many financial institutions. MOVEit Transfer had a critical SQL injection vulnerability CVE 2023 34362 that hackers exploited before a patch was available, stealing data from organizations using it.
- Several banks had to disclose data breaches as a result, even though their own systems weren't flawed it was a third party software.
- This trend continued into 2025: attackers look for any software common in finance trading platforms, core banking software, etc. and probe for vulns.

- **Defensive posture:** Financial orgs are increasingly doing **continuous penetration testing** and **red team exercises** to find weaknesses before criminals do.
- Many are also investing in **attack surface monitoring** because large banks have sprawling infrastructure and may not realize a forgotten server or app is exposed with a known vuln.
- Regulatory pressure is mounting too: regulators now often ask for evidence of vulnerability management programs and even scenario testing e.g., OCC in the US expects banks to address critical vulns swiftly.
- The sector has improved cyber defenses, but the **threat level remains extreme**. With the average cost of a financial breach around $5.9M, the incentive to shore up vulns is clear.

IBM's 2023 Cost of a Data Breach report found the **average breach cost for financial services was $5.56M**, second only to healthcare. And for the first time, the global average cost per breach actually dipped slightly to $4.8M in 2024 from $4.45M, possibly due to faster response and containment. Still, breaches in finance often cost well above that average.

Healthcare: Hospitals & Medical Organizations

The healthcare sector has been **hit hardest by cyber attacks** recently, and 2025 is no exception. In 2023, healthcare providers in the U.S. suffered **809 breaches**, up **136%** from 2022 the most of any industry.

Over **56 million patient records** were compromised in those incidents, exposing everything from medical histories to Social Security numbers.

Healthcare breaches can be devastating not just due to data loss but because they can disrupt patient care e.g., hospital ransomware forcing diversion of ambulances.

- **Shift to hacking:** Historically, a lot of healthcare breaches were caused by human errors, lost laptops, mis mailed records. But now, **malicious attacks dominate**.
- By 2024, **67% of healthcare breaches were attributed to external attackers** hackers/ransomware, while only 30% were internal accidents or insiders.
- This is a big change it means cybercriminals are actively targeting hospitals and clinics, likely because medical data is lucrative and hospitals often have weaker security.
- The rise in **espionage motivated breaches** is notable too: there have been cases of state sponsored actors targeting biotech and vaccine research via exploiting vulnerabilities in healthcare systems DBIR 2025 flagged more espionage in healthcare than prior years.

Common vulnerabilities/exploits:

- Healthcare networks are often a mix of old and new legacy systems sometimes running outdated OS alongside modern IoT medical devices.
- Attackers exploit unpatched **VPN servers, RDP endpoints, or database software** in healthcare environments to gain entry.
- For instance, the 2023 ransomware attack on a hospital chain was traced to an **unpatched VPN appliance with a known CVE** that allowed remote access. Once in, attackers deploy ransomware, crippling systems.

- Ransomware is **endemic in healthcare** now in Verizon's data, healthcare orgs experienced a huge surge in ransomware incidents.
- In fact, **88% of breaches at small healthcare orgs involved ransomware** in the past year, smaller clinics being easier prey, compared to 44% at larger ones.
- Another major risk is outdated medical devices, MRI machines, infusion pumps running Windows 7 or older they often have known vulns but can't be easily patched due to FDA regulations or uptime needs. Attackers have exploited such devices as pivot points.

Impact and response:

- The average cost of a healthcare breach hit **an all time high of $10.93M** in recent IBM research, the highest of any industry for the 13th year in a row. Apart from fines, HIPAA penalties and lawsuits, breaches can literally put lives at risk if systems for patient care go down.
- Regulators are pressing hard: the U.S. HIPAA enforcers fine entities for not addressing known risks, and new guidance from HHS 405d calls for improved vulnerability management in health IT.
- Hospitals are trying to implement network segmentation to isolate medical devices and **24/7 monitoring** to catch intrusion signs early. There's also a push for **penetration testing in healthcare** though not explicitly mandated by HIPAA, it's implied under risk assessments.
- Many healthcare orgs now run periodic **vulnerability assessments** on their networks and medical device inventories, to prioritize patching of, say, a known Windows server bug that could lead to a breach.

Healthcare faces a perfect storm valuable data, sometimes lax security, and a life or death imperative to stay online. Attackers know many hospitals will pay ransom to restore operations quickly, so they aggressively target any vulnerability that can get them in.

From a **vulnerability statistics** perspective, healthcare orgs should pay special attention to **phishing defenses** since human error still plays a role and to **patching external facing systems** VPNs, portals.

The days of relying on perimeter firewalls alone are over, now it's assumed the perimeter will be breached via an unpatched vuln, and you need layers of defense beyond that.

Government & Public Sector

Government agencies federal, state, local are in the crosshairs of both cybercriminals and nation state hackers. While the total number of public sector breaches is lower than industries like healthcare, the **impact can be very high, think** nation state espionage or critical infrastructure disruption.

In 2023, the U.S. government sector reported **100 breaches**, a **35% increase** from 2022, affecting around **15 million individuals** mostly in large incidents. These include everything from local city hacks to major federal agency compromises.

APT exploits:

- Nation state APT groups frequently exploit **zero days and known vulns** to infiltrate government networks.
- A notable example was in 2023 when multiple U.S. federal agencies were breached: a Chinese state sponsored group exploited a new flaw in Microsoft Exchange Server and a vulnerability in a Cisco router firmware to gain persistent access.
- This multi pronged attack showed how determined adversaries will chain exploits.
- Another incident involved a years old vulnerability in a file transfer app that a foreign actor used to steal data from a government database embarrassing because the vuln had been known and patchable.

Even older CVEs remain relevant in gov targets. In 2025, CISA added a **2010 era Internet Explorer bug CVE 2010 3962** to its exploited catalog because it was observed being used by a sophisticated group likely repurposed from older cyber weapons.

This highlights that some government systems might still be running legacy software, and adversaries will dust off decade old exploits if they work.

- **Vulnerabilities and mandates:** Governments have responded by getting more aggressive on vulnerability management. The U.S.

Cybersecurity and Infrastructure Security Agency CISA issues **Binding Operational Directives BODs** that mandate federal agencies to patch certain CVEs by set deadlines.

- For example, BOD 22 01 established a constantly updated list of Known Exploited Vulns that agencies must remediate, often within 2 3 weeks of listing. Fail to patch by the due date, and you're out of compliance with potential repercussions.
- As of October 2025, this list includes hundreds of CVEs across Windows, Cisco, Oracle, you name it. The feds even require agencies to **track and report** their progress on these patches.
- This has significantly improved patch cadence in some areas. State and local governments, while not under federal BODs, are encouraged to follow suit.

The EU's **NIS2 Directive** and similar laws worldwide are pushing government entities to establish better vulnerability disclosure and management. The EU in 2025 launched the **European Vulnerability Database EUVD** to centralize info on vulns affecting the EU market.

ENISA the EU cybersecurity agency became a CVE Numbering Authority to help coordinate vuln disclosures in Europe. This reflects the global recognition that governments need to know about and fix vulns faster.

Threat landscape:

- Government networks are attractive to spies, so zero days are often used e.g., Exchange Server 0 days in 2021, Pulse Secure VPN 0 days in 2021 targeting agencies, etc.. But also ransomware groups hit local governments hard city governments have been extorted via known vulns in their systems.
- The mix of outdated tech and sensitive data can be problematic. For instance, many government offices still run legacy apps that only work in IE or old Java creating a field day for CVEs.

Securing public sector:

- Governments are investing in things like **bug bounty programs** the DoD's Hack the Pentagon was an early example to uncover vulns.
- They're also increasingly requiring contractors to attest to vulnerability management practices e.g., the U.S. DoD's CMMC program touches on keeping software updated.

- **Supply chain security** is a big concern after the SolarWinds incident which itself was a vulnerability in the software build process exploited by Russia in 2020. Now agencies ask for SBOMs Software Bill of Materials to track components and their known vulns.

In summary, the public sector has improved structure with CISA directives, etc., but it remains a top target. The **risk of unpatched vulns** leading to a breach is something agency heads are very aware of now likely spurring increased budgets for modernization to get off that Windows 2008 server vulnerable to EternalBlue, finally!.

Bright spot: So far, no catastrophic cyber attack on critical infrastructure via a vulnerability has happened in 2025, but experts warn it's possible e.g., a power grid system with an unpatched bug could be a target.

Thus, governments worldwide are practicing cyber war games and drilling responses to major vuln exploits in critical systems. Preparedness is improving, but the threats are evolving too.

Retail & E Commerce

Retailers and e-commerce companies continue to face cyber threats, though they often get less press than finance or healthcare.

In 2023, the retail sector had **119 reported breaches**, exposing about **10 million customer records**. The counts are lower, but remember many small retailers might not disclose or even detect breaches, so the real number could be higher.

Attacks on retailers:

- A lot of retail breaches center on **payment data theft**. Traditional attacks included **point of sale POS malware** like the breaches of Target, Home Depot back in the day. As chip and pin cards made POS hacks harder, attackers shifted online to e-commerce skimming.
- In 2025, **Magecart style web skimming** is still rampant: attackers inject malicious JS into checkout pages to steal credit card details in real time. What's new is how sneaky these have become as mentioned earlier, scripts that only activate during the final checkout to avoid detection by security scans.

- A single vulnerability in a retailer's web platform, say, an **outdated plugin on a Magento or WooCommerce site** can allow such injection. There was a case where a vulnerability in a third party chat widget script let attackers compromise hundreds of online stores with Magecart code.

Retailers also face a ton of **credential stuffing** attackers trying leaked passwords on e-commerce accounts, but that's not a vulnerability in code though some have had vulnerabilities in their APIs that made such attacks easier. For instance, an API with no rate limit could let attackers test thousands of logins quickly.

SMBs at risk:

- Many retailers are small to mid businesses with limited IT security staff. They may not promptly patch their e-commerce software or update that one Linux server in the back office. Attackers know this and run broad scans.
- A common scenario: an attacker scans for a known vuln in Magento, a popular online store platform, finds 50 stores still unpatched, injects a web shell and skimmer code voila, credit card data begins streaming out.
- The retailers often only notice when customers report fraud or an external researcher finds the malicious script.

- **Regulatory environment:** The payment card industry's standards PCI DSS have forced larger retailers to secure their systems e.g., requiring regular scans and **penetration testing** for Level 1 merchants.
- Compliance has improved things somewhat it's no longer as easy to hit a Fortune 500 retailer with a basic SQLi and dump millions of card numbers, PCI requires segmentation and encryption, etc..
- However, **smaller merchants** not under heavy PCI scrutiny might be more lax. Also, retailers have to manage third party risk: many outsource parts of their ecommerce or have numerous integrations inventory systems, payment gateways. Any of those could introduce vulns.

Recent trend:

- Ransomware crews started going after retailers too, especially targeting the back end systems like **inventory management or supply chain systems**.

- In late 2024, a major supermarket chain was hit by ransomware that exploited a known vulnerability in their VPN appliance. The ransom note basically said pay up or we'll keep your stores offline.
- This shows how a vuln in an IT system can translate to tangible losses stores unable to process transactions.

Defenses for retail:

- For e commerce sites, keeping the platform and plugins updated is rule #1. Many are also deploying **web application firewalls WAFs** to catch common attacks though WAFs can be bypassed if misconfigured.
- Retail IT teams, even small ones, should at minimum run automated **vulnerability scans** regularly many web hosts offer scanning for their customers now.
- Training is also needed: e.g., store employees should be wary of phishing there have been cases where attackers phish retail employees to get access to admin panels, then exploit a vuln further.

A positive development is more retailers using **managed security services** since they can't afford full time experts. These services can monitor their site and network for known exploits 24/7.

Also, **Penetration Testing as a Service PTaaS** is becoming popular: instead of a one time test, it's an ongoing subscription where testers continually probe the retailer's systems for new vulns. This can be a lifesaver given how quickly new CVEs appear.

Technology & Software Companies

It's ironic, but technology companies including software vendors, SaaS providers, etc. often find themselves victims of the very vulnerabilities they track.

In 2023, the tech sector had about **167 breaches** reported, impacting **65 million records** likely skewed by a few big ones like maybe a large tech firm breach.

Supply chain attacks:

- One of the biggest threat trends for tech companies is **software supply chain attacks**. These target the development pipeline or update process.
- In 2025, there was reportedly a **22% increase in supply chain attacks on open source software components**, as more attackers sneak malicious code into libraries that tech firms consume. For example, malicious NPM packages and typosquatting incidents skyrocketed.
- A single compromised library can infect thousands of downstream products, recall the **Log4j vulnerability Log4Shell** in late 2021 which affected countless organizations.
- In 2025, no single vuln has had that scale yet, but there have been many mini Log4Shells where a vulnerability in a common component say OpenSSL or a popular NPM module forces emergency patching across the industry.
- We also saw **breaches of tech company CI/CD systems**: in one case, attackers exploited a vulnerability in a DevOps tool to get into a software company's network, then stole source code which could later be analyzed for further vulns.
- The infamous **SolarWinds attack** of 2020 where Russian hackers injected a backdoor via a build system vuln still looms in memory, it has spurred tech firms to double check the security of their build and update processes.

Bug bounty data and zero days:

- Tech companies run bug bounty programs, which means they probably know about more vulnerabilities via private reports than anyone. The positive is they can patch those before public disclosure.
- The negative is, if a bounty program isn't well run, a reported vuln could leak or not get fixed quickly.
- Luckily, most large tech firms Google, Microsoft, etc. are pretty efficient at handling vulns. Smaller startups though might struggle if a researcher drops a 0 day on Twitter about their product.

Insider and misuse:

- Another aspect in tech breaches is that sometimes **insider mistakes or misconfigurations** lead to exposure.

- For example, cloud storage buckets left open not a software vuln per se, but a human vuln. However, even that is being addressed by cloud security tools now that flag open buckets.

Response in tech sector:

- Being security forward, many tech companies have embraced frameworks like **NIST CSF 2.0** and **ISO 27001**.
- The updated ISO 27001:2022, for instance, has a specific control Annex A 8.8 requiring a formal **vulnerability management process**. So tech firms aiming for ISO certification have to show they identify, assess, and treat vulns in a timely way.
- Additionally, most reputable software companies now issue **security advisories** and publish CVEs for their products transparency has improved. Microsoft's Patch Tuesday, Adobe's security bulletins, etc., are examples of how the tech sector handles disclosing and fixing vulns on a routine schedule.

Another initiative gaining traction is **SBOM Software Bill of Materials** usage. Some tech companies provide SBOMs for their products so customers know if, say, a given version includes Log4j and needs patching when Log4j has a new CVE.

This helps propagate vulnerability info quickly through the user base. The U.S. government is pushing SBOM requirements for software sold to agencies, which affects many tech vendors.

Finally, tech companies are frequently **penetration testing their products** some even offer pentest reports to customers as proof of security. Cloud providers like AWS, Azure have whole teams doing this constantly internally.

There's also collaboration via **platforms like HackerOne** where multiple tech companies share vulnerability data.

In summary, the tech sector, while more mature in security, is not immune to big breaches. If anything, an exploited vulnerability in a tech company's product can cascade to thousands of other businesses supply chain effect. So stakes are high.

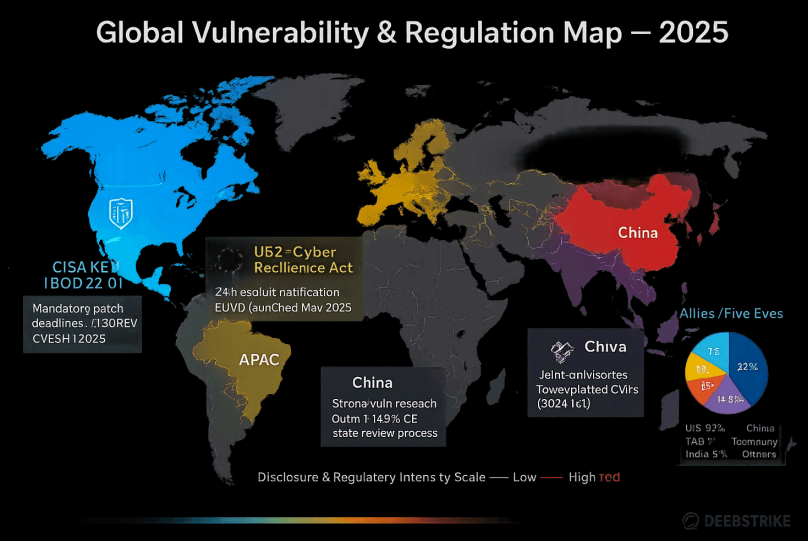
The flip side is, tech companies often lead the way in adopting new security practices like **Zero Trust Architecture** and automated **code scanning SAST** to catch vulns early. We can expect them to

continue hardening their development lifecycles to reduce vulnerabilities from the get go.

The big picture across industries:

- Virtually every sector is seeing **more vulnerabilities and more aggressive exploitation**. Whether it's a hospital or a bank or a government agency, the story is similar: if you don't patch your known flaws, you're at high risk of an incident.
- Attackers will adjust their tactics depending on target phish more in healthcare, exploit web apps more in retail, etc., but the underlying theme is unpatched vulnerabilities remain one of the **leading causes of breaches** across the board.
- According to Verizon DBIR 2025, vulnerability exploitation was involved in 20% of breaches, a huge rise and phishing in 16%, meaning technical vulns are now surpassing some social engineering in frequency.

Geographic Trends & Regulations in 2025



Cyber vulnerabilities and their management have become a **global concern**, with different regions responding via new laws and collaborative efforts.

Where do vulnerabilities originate? In terms of sheer numbers of CVEs, some analyses tie them to countries based on where the software/vendor is or researcher reporting. The U.S. and China are top sources of vulnerabilities in CVE databases.

One mid 2025 analysis suggested the U.S. was associated with 32% of CVEs over 9,300 and China 14.6% about 4,200. This reflects active security research and lots of software in those countries.

Other countries like Germany, India, Japan, South Korea each accounted for 5-7% of CVEs. It's interesting because it shows vulnerability discovery is worldwide.

The CVE program has nearly 500 partner organizations CNAs across 40+ countries issuing CVEs now. So, the effort to find and catalog vulnerabilities is very global.

North America U.S. focus:

- As we covered, the U.S. has taken a hard stance on known vulns with CISA's directives. After some big incidents Colonial Pipeline ransomware, etc., the government realizes unpatched vulns are a national security threat.
- The U.S. SEC even passed a rule in 2023 requiring public companies to disclose material cyber incidents indirectly forcing them to be more upfront if a breach happened due to, say, an unpatched vuln.
- We're also seeing insurance companies in the U.S. scrutinize clients' patch management, some cyber insurance policies might not pay out if a breach was from a known vulnerability that wasn't patched in a reasonable time.
- This is making corporate boards pay attention to vulnerability stats.

Canada is similar in adopting NIST based frameworks and encouraging patch discipline, but hasn't gone as far with mandates as the U.S.

Europe EU:

- The EU's **NIS2 Directive** came into force, expanding the scope of companies that must follow cybersecurity risk management including vulnerability handling and report incidents.
- Under NIS2, many more sectors like medium businesses in healthcare, finance, cloud providers, etc. must take vulns seriously or face fines. A tangible outcome: ENISA launching the **EU Vulnerability Database EUVD** in May 2025.

- The EUVD aggregates info from CVE, vendors, CSIRTs, etc., and crucially includes whether a vuln is being exploited and any mitigations. It essentially complements the U.S. NVD/KEV with a European flavor. ENISA also became a CNA to assign CVEs for vulns discovered by EU bodies.
- And looking forward, the **Cyber Resilience Act CRA** will, by 2026, require manufacturers of digital products to fix and **notify authorities of any actively exploited vulnerability within 24 hours**.
- Yes, 24 hours! That's a stringent rule meaning if you sell software/hardware in EU and you find a vuln that's being exploited in the wild, you must inform regulators within one day and patch asap. This aims to prevent cover ups and speed up protective actions for users.

Europe also saw its share of attacks: an ENISA report noted that in early 2025, **21.3% of incidents in Europe were traced to vulnerability exploitation** phishing was 60%. So Europe's threat landscape is similar.

They also had big cases like the **MoveIT breach** hitting British companies, etc. Therefore, EU is doubling down on coordinated vulnerability disclosure and patch enforcement.

Asia Pacific:

- APAC is diverse, with countries like Singapore and Australia being quite proactive both have breach notification laws and national cyber strategies that include vulnerability management.
- Australia in 2023 started an initiative for a **national coordinated vulnerability disclosure platform** so researchers can report vulns in Australian critical infrastructure.
- India's CERT In issued rules in 2022 requiring organizations to report cyber incidents including if a vulnerability was likely involved within 6 hours.
- Japan has a well oiled vul disclosure process via JPCERT. However, many APAC organizations still lag in patching often due to resource constraints or reliance on legacy tech.
- Attackers exploited this: e.g., some 2024 25 campaigns by Chinese APTs targeted Southeast Asian telcos via old vulnerabilities in routers that hadn't been updated, granting them spy access.

One notable APAC event **the Red Cross data breach in the Asia region 2022** happened due to an unpatched critical vuln in an Adobe ColdFusion server. That reverberated across humanitarian sector. By 2025, even NGOs in APAC are trying to get better at patching.

Global cooperation:

- The U.S. and allies have started sharing more vulnerability info. For instance, the Five Eyes nations frequently release joint advisories listing the top exploited CVEs in the past year, a helpful heads up to all sectors to patch those if they haven't.
- A joint advisory in 2025 listed CVEs in Fortinet, Citrix, and Exchange as the most exploited over 2024, which lined up with what we saw.

Key regulatory trends to remember:

- **Mandatory patching deadlines:** e.g., U.S. federal agencies must patch KEV listed vulns by X date. Expect this approach to trickle to other sectors perhaps via insurance or sector specific regs.
- **Mandatory vulnerability disclosure by vendors:** e.g., EU CRA 24 hour rule, and similar laws are being discussed in places like Singapore.
- **Reporting of vulnerabilities:** NIS2 requires EU countries to set up processes to handle vulnerability info meaning companies should have a plan to report and receive reports. The U.S. already has some of this like FDA guidance that medical device manufacturers must report certain vulns.
- **Liability shift:** There's talk especially in EU about holding software makers more liable for security issues. This is controversial, but if it happens, it could drastically change how vulns are handled. Maybe companies will invest more in pre release security to avoid liabilities.

In conclusion on the geo/regulatory front: **vulnerability management is now a legal expectation, not just IT best practice.** Organizations that ignore critical updates risk not only breaches but also fines and legal trouble.

The upside is governments and industry groups are providing more resources like the KEV catalog, EUVD, and threat intel to help prioritize the most dangerous vulnerabilities first. In 2025 and

beyond, staying ahead of vulnerabilities will be a key part of staying compliant and cyber resilient.

How to Keep Up: Best Practices for Vulnerability Management in 2025

With the torrent of vulnerabilities and exploits, organizations need a solid game plan to stay secure. Here's a **5 step approach** to effective vulnerability management VM in the current landscape:

Inventory Your Assets & Software:

- You can't secure what you don't know about. Maintain an up to date inventory of all hardware and software in your environment, including versions. Use automated discovery tools to find shadow IT or forgotten systems.
- This inventory forms the basis for knowing which CVEs apply to you. For instance, if OpenSSL releases a security advisory, you should instantly know which servers use OpenSSL and need patching.
- Many breaches occur on systems that IT didn't even realize were exposed with vulnerable software.

Continuously Monitor for Vulnerabilities:

- Gone are the days of annual or quarterly scans you need continuous monitoring. Implement regular automated scans using tools like **Nessus, Qualys, OpenVAS** etc. across your network to identify missing patches or misconfigurations.

- Additionally, subscribe to threat intelligence feeds and vulnerability alert services like CISA alerts, vendor bulletins.
- Modern VM tools can correlate your asset inventory with the latest CVE disclosures to flag Hey, you have Tomcat 9, and a new CVE just came out for it today.
- Consider using **attack surface management** services that monitor your internet facing assets from an attacker's view they can often spot an outdated system you forgot about. And don't ignore application code: incorporate **SAST/DAST** scanning in your dev pipeline to catch vulnerabilities before deployment.

Prioritize Based on Risk:

- With hundreds of new vulns monthly, prioritization is key. Don't just rely on CVSS scores, consider context. A critical CVE on a system that's isolated behind multiple layers might be less urgent than a Medium vuln on an exposed web server.
- Use frameworks like **CVSS + EPSS Exploit Prediction Scoring System** and CISA's KEV list to gauge likelihood of exploit. Many VM programs now use **risk based scoring** factoring in whether exploit code is available, whether the asset is internet accessible, and the sensitivity of data on it.
- For example, a 7.5 CVSS bug that's already actively exploited in the wild gets top priority. Some tools integrate threat intel so you know, say, a ransomware gang is leveraging that vulnerability meaning patch it yesterday.
- Create a heatmap of vulns and focus efforts on the critical few the 20% of issues that pose 80% of the risk.

Rapid Patching & Mitigation:

- Develop a streamlined patch management process. This means having maintenance windows or automated patch deployment for various systems so that applying updates is routine, not disruptive.
- For critical vulns, you might need out of cycle emergency patching. Aim to shrink your mean time to patch for high severity issues to days, not weeks the industry average was around 30 days for edge devices try to beat that.
- When you can't patch immediately maybe the fix isn't out or it's a legacy system, implement mitigations: e.g., temporary firewall rules, shutting off a service, applying vendor workarounds.

- Why this matter? We saw that if you can patch critical vulns within 48 72 hours, you'll avoid most mass exploitation, since many attacks occur within a week of disclosure. It's tough, but that's the goal.
- Utilize deployment tools to push patches overnight. And don't forget to patch third party software and libraries included in your applications supply chain vulns. Using a **software composition analysis** tool can help track those.

Verify and Repeat:

- After patching or mitigation, verify that the vulnerability is indeed closed. This could involve rescanning the system or even doing a targeted **penetration test** to ensure the exploit no longer works.
- Many organizations use **Penetration Testing as a Service PTaaS** platforms that continuously test key assets this acts as a second check that critical vulns are squashed and no new ones have opened up.
- Also, incorporate lessons learned: if you find that a critical vuln went unpatched because a system was off the radar, update your inventory process.
- Establish KPIs for your VM program e.g., % of critical vulns patched within SLA and report to management.
- Regularly schedule a review say monthly of outstanding vulns and progress.
- Essentially, vulnerability management is a continuous loop, feeding back into asset management and threat monitoring.

The fix is in the follow through: A shiny vulnerability scanner tool means little if you don't act on it. Make sure there's accountability assign owners for remediating issues. Build a culture where ops teams and security work together, not at odds, to get patches deployed safely.

Where possible, **automate** patching of lower environments and even production if risk is acceptable some cloud native orgs auto update minor patches.

Use maintenance windows smartly to avoid downtime excuses. In some cases, consider **virtual patching** via WAF rules or IPS signatures that can block exploit attempts until you patch the actual system.

By implementing these steps, organizations can drastically reduce their exposure. Remember that a **majority of attacks still leverage known vulnerabilities** rather than 0 days. Simply keeping up with patches and basic hardening could prevent an estimated 80-90% of attacks. It's not glamorous, but it works.

The vulnerability landscape in 2025 paints a clear picture: **cybersecurity vulnerabilities are more numerous, more quickly exploited, and more consequential than ever before**. Organizations are facing an onslaught of new CVEs, likely around 50,000 this year with a significant portion being high severity.

Attackers, from ransomware gangs to state sponsored APTs, have honed their ability to weaponize these flaws at lightning speed, often within hours of disclosure. It's no wonder that vulnerability exploitation now accounts for **20%+ of breaches**, rivaling phishing and stolen creds as a top attack vector.

The data and trends we've explored highlight a few key takeaways:

Managing vulnerabilities is a strategic imperative:

- This is not just an IT problem, it's a business risk and often a compliance requirement.
- Leadership needs to invest in the people and tools for effective vulnerability management, because the cost of not doing so breaches, downtime, reputational damage far exceeds the investment.
- The threats of 2025 demand more than ad hoc patching, they require a mature, continuous process.

Prioritization and speed are everything:

- You will never patch everything immediately nor do you need to. The organizations that fare best are those that can rapidly identify which vulnerabilities pose the greatest risk e.g., known exploited, critical systems exposed and remediate those first.
- Being able to patch critical vulns in days not months is what separates companies that avoid incidents from those that fall victim. Aim to shrink that window of exposure relentlessly.

Defense in depth is crucial because some vulns will slip through:

- Given the sheer volume, it's likely something, somewhere in your stack, is vulnerable at any given moment. That's why multiple layers of security controls are needed.
- Network segmentation, up to date intrusion prevention systems, application firewalls, endpoint detection & response these can catch or mitigate exploit attempts even if a vuln exists.
- Regular **penetration testing** and red teaming can also help by identifying holes in your defenses before attackers do.
- Remember, **testing isn't a one time event**, with continuous integration/deployment, new vulns can be introduced quickly, so testing must keep pace.

Compliance and regulations are tightening:

- From CISA's KEV mandate to the upcoming EU CRA law, the message is clear organizations will be held accountable for known vulnerabilities.
- Demonstrating strong **E-E-A-T Experience, Expertise, Authoritativeness, Trustworthiness** in security, including how you handle vulns, is becoming part of due diligence in many industries even impacting cyber insurance premiums and contracts.
- Keeping an eye on regulatory changes and aligning your vulnerability management to them will save you pain down the road.

Everyone has a role to play:

- Ultimately, reducing vulnerabilities is a team effort. Developers need to code more securely shout out to practices like threat modeling and secure code training to cut down those OWASP Top 10 issues. IT ops need to embrace automation and not fear patches.
- Security teams need to communicate risk in terms the business understands e.g., this vuln could enable a ransomware attack that halts production.
- And end users should be educated too many vulnerabilities like phishing or social engineering take advantage of human factors. A holistic approach covers people, processes, and technology.

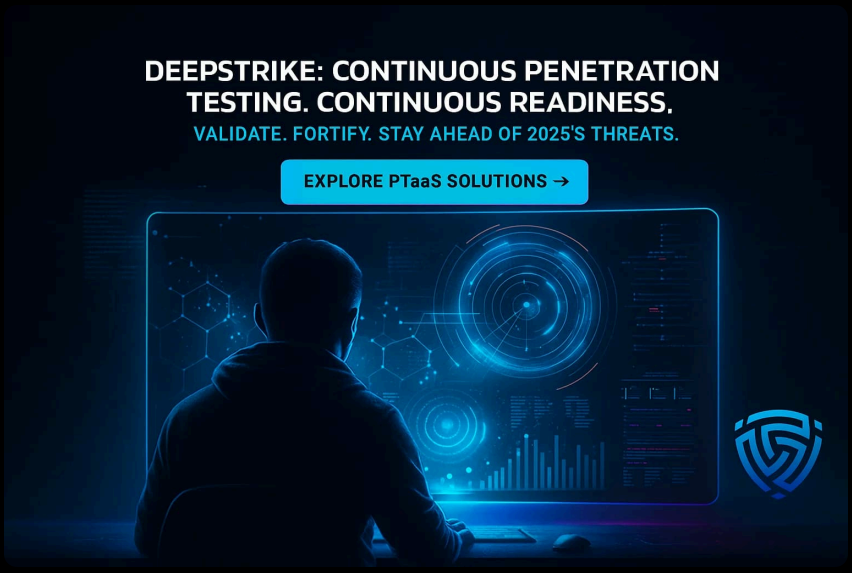
If there's a silver lining, it's that we have more tools and knowledge than ever to combat this. The fact that we can track 50k vulns a year

is itself progress thanks to global collaboration. Initiatives like KEV and EUVD make it easier to focus on what matters. And the cybersecurity community is quick to share information on new threats, the rapid mobilization around Log4Shell in Dec 2021, for instance, showed how fast defenders can move when alerted.

As we move beyond 2025, one can expect **AI to play a bigger role** both for attackers perhaps automating vuln discovery and for defenders automating patch management or predicting which vulns will be exploited next. The cat and mouse game will continue. But with a solid foundation in vulnerability management, organizations can dramatically tilt the odds in their favor.

Ready to Strengthen Your Defenses?

The threats of 2025 demand more than just awareness, they require readiness. If you're looking to validate your security posture, identify hidden risks, or build a resilient defense strategy, DeepStrike is one of the **top penetration testing companies** is here to help. Our team of practitioners provides clear, actionable guidance to protect your business.



Explore our **penetration testing services** to see how we can uncover vulnerabilities before attackers do. **Drop us a line** we're always ready to dive in and fortify your defenses.

About the Author

Mohammed Khalil is a Cybersecurity Architect at DeepStrike, specializing in advanced penetration testing and offensive security operations. With certifications including CISSP, OSCP, and OSWE,

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Frequently Asked Questions FAQs

How many CVEs have been published in 2025 so far?

- As of mid 2025, around **23,667 CVEs** were published in the first half of the year. That's roughly a 16% increase over H1 2024.
- If the trend continues, the total CVEs for 2025 might approach **50,000** by year's end, a new record.
- On average, we're seeing about **130 new vulnerabilities disclosed per day** in 2025.

Are vulnerabilities getting more severe, or just more numerous?

- Both. The volume of CVEs is increasing, and a significant chunk are high severity.
- In 2025 to date, roughly **38%** of reported vulns are rated **High or Critical severity** CVSS ≥7. Early 2025 actually saw a spike in Critical CVSS 9+ vulns compared to prior years.
- So we have more vulnerabilities overall, and many of them 4 in 10 pose serious risk. That said, not every CVE is actually exploitable or impactful in real environments but plenty are.

What are the most commonly exploited vulnerabilities in 2025?

- Attackers love to hit known, unpatched flaws. Some of the top exploited CVEs in early 2025 include vulnerabilities in **Microsoft products**, various **VPN appliances Fortinet, Ivanti, Citrix**, and popular web software.
- For example, a specific Ivanti Pulse Connect Secure VPN bug CVE 2025 0282 and a Fortinet FortiOS bug were heavily abused by state actors. Also, older flaws like **Log4Shell CVE 2021 44228** and **ProxyShell Exchange** are still being exploited in unpatched systems.
- CISA's KEV Catalog is a good reference, it lists hundreds of actively exploited CVEs, many of which are from 2017-2022 but

attackers still find victims.

- In summary: unpatched known vulns in perimeter devices and Windows systems remain top targets.
- Additionally, any new critical vuln that gets a public exploit PoC tends to see immediate exploitation e.g., the MOVEit Transfer zero day in mid 2023 was one of the most exploited of 2023 and continued into 2024.

How fast do attackers exploit new vulnerabilities now?

- Extremely fast, often within **24 to 48 hours** after a vulnerability's details go public.
- In Q1 2025, roughly **28%** of exploits were observed within one day of CVE disclosure. Sometimes we even see pre patch exploits true 0 days.
- In general, for a critical vulnerability, assume attackers will start scanning and attacking immediately.
- We've seen cases like the April 2024 Barracuda Email Appliance vuln exploit attempts began the same day the advisory was announced.
- The era of patch available, but we can wait a few weeks is over, that's how organizations get hit by day exploits quickly.

What's the difference between a vulnerability assessment and a penetration test?

- A **vulnerability assessment** is an automated scan using tools that identifies known vulnerabilities in systems, essentially generating a list of these systems are missing patches or have config weaknesses. It's broad but shallow no active exploitation.
- A **penetration test** pentest, on the other hand, is a deeper manual engagement by ethical hackers attempting to actually exploit vulnerabilities and security gaps in an environment, often chaining them together like a real attacker would.
- Pentesting can find complex issues, logic flaws, chained attacks that scanners might not catch, and it provides proof of concept of impact.
- Think of vulnerability assessment as vulnerability detection, whereas penetration testing is vulnerability validation and exploitation.

- Both are important: typically you'd run regular vuln scans to manage routine patching, and conduct pen tests periodically or continuously via a **PTaaS** platform to uncover deeper security issues and to test your defenses.
- For more details, check out our guide on **vulnerability assessment vs penetration testing** which dives into use cases for each.

What is CISA's Known Exploited Vulnerabilities KEV catalog?

- CISA's KEV Catalog is a curated list of CVEs that are known to be actively exploited in the wild . It was launched in late 2021 and updated continuously.
- When attackers are observed using a vulnerability per threat intel/government agencies, CISA adds that CVE to the catalog. U.S. federal agencies are required by policy Binding Operational Directive 22 01 to **patch every KEV listed vuln by a specified due date**.
- The KEV list is public and very useful for all organizations it's essentially a patch these first list. As of 2025 it contains hundreds of vulns, from old ones like CVE 2010 3962 IE bug to very recent ones.
- Using KEV, you can prioritize fixing those vulnerabilities that are confirmed in active attacks, which gives you a leg up in risk reduction.

How much does a penetration test cost in 2025?

- The cost of a penetration test in 2025 varies widely depending on scope, depth, and provider.
- A simple web application pentest might range from **$5,000 to $15,000**, whereas a comprehensive network and application pentest for a large enterprise could be **$50,000+**.
- Factors include whether it's black box vs white box white box can be more efficient testing with access, the number of IPs/apps in scope, and the experience level of the testing team.
- Some companies opt for **continuous penetration testing or PTaaS** subscriptions which might be, say, $3k- $10k per month but cover multiple iterative tests and re testing.
- While it's an investment, consider that the **cost of a data breach** in 2025 averages **$4- 5 million** globally, spending a fraction of

that on proactive testing is often worth it. Also, compliance requirements PCI DSS, SOC 2, etc. might dictate annual or quarterly pentests, so budget accordingly.

- For more detail, see our breakdown of **Penetration Testing Cost in 2025**, where we compare black box vs white box testing costs and typical pricing models for web app vs network vs cloud pentests.